

**Wonder Woman 2 filming**

The film will be set during the cold war and will be mostly shot in America, with Director Patty Jenkins back at the helm. <http://digit.in/WW2USA>

**YouTube gun ban**

YouTube bans video promoting gun modifications, tutorials and hacks. <http://digit.in/gunban>



COMPUTER VISION

Tech that gives computers eyes is the need of the hour and you could be a part of this field.

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hen you look around, there are more things looking back at you today than ever before - most of them

non-living. Before this intro creeps you out, let us explain that this is about computer vision. Or more specifically, how to build a career in this area.

Creepy intro apart, it is indeed true that thanks to fields like autonomous driving, augmented reality, demand for specialised computer vision skills in the industry is growing at an unprecedented rate. So, if you ever fancied giving computers and gadgets their own pairs of eyes, now is the right time to dive deep into the field of computer vision.

WHAT IS IT?

The term 'computer vision' is broad, encompassing a number of technology areas that work together to make the idea of a computer having eyes feasible. With the objective to allow artificial systems to dynamically gather data from their surroundings, the area



Some of the best companies in the world are trying to develop self driving technology, a field that heavily relies on computer vision

can be broadly divided into a number of basic specialisations:

- Fundamentals of image formation
- Camera imaging geometry
- Feature detection and matching
- Stereo, motion estimation and tracking
- Image classification and scene analysis

When it comes to specific applications, skills like image recognition, camera calibration, are also required for areas like surveillance and self driving cars. While its applications might be relatively new, computer vision itself is evolving in a new direction and is adopting changes in its basic methodologies as well. "In almost all use cases, the deep learning algorithms outperform traditional computer vision algorithms; hence the industry is moving towards deep learning based approaches to computer vision," says Ankur Pandey, Sr. Data Scientist, HERE Technologies. "On the hardware end, the focus is

on on-premise computations - which means creating hardware which can do computer vision on the spot on the embedded machine itself, without using the cloud."

THE SKILLSET

Just like the applications of computer vision, the skills required to work with them are diverse and range across a few subjects and areas. One of the first things that such jobs entail is a good grasp over mathematics - especially topics like linear algebra and vector calculus. In addition to that, understanding data structures and the representation of images, features and geometric constructions through data is necessary to work with computer vision. When it comes to tools, Python with NumPy is a prerequisite even in the online courses that will teach you about computer vision. Although it is a bit dated now, Matlab is still a very capable and versatile tool when it comes to computer vision and image processing, and a proper knowhow of



Python with NumPy is essential in almost all kinds of computer vision roles.